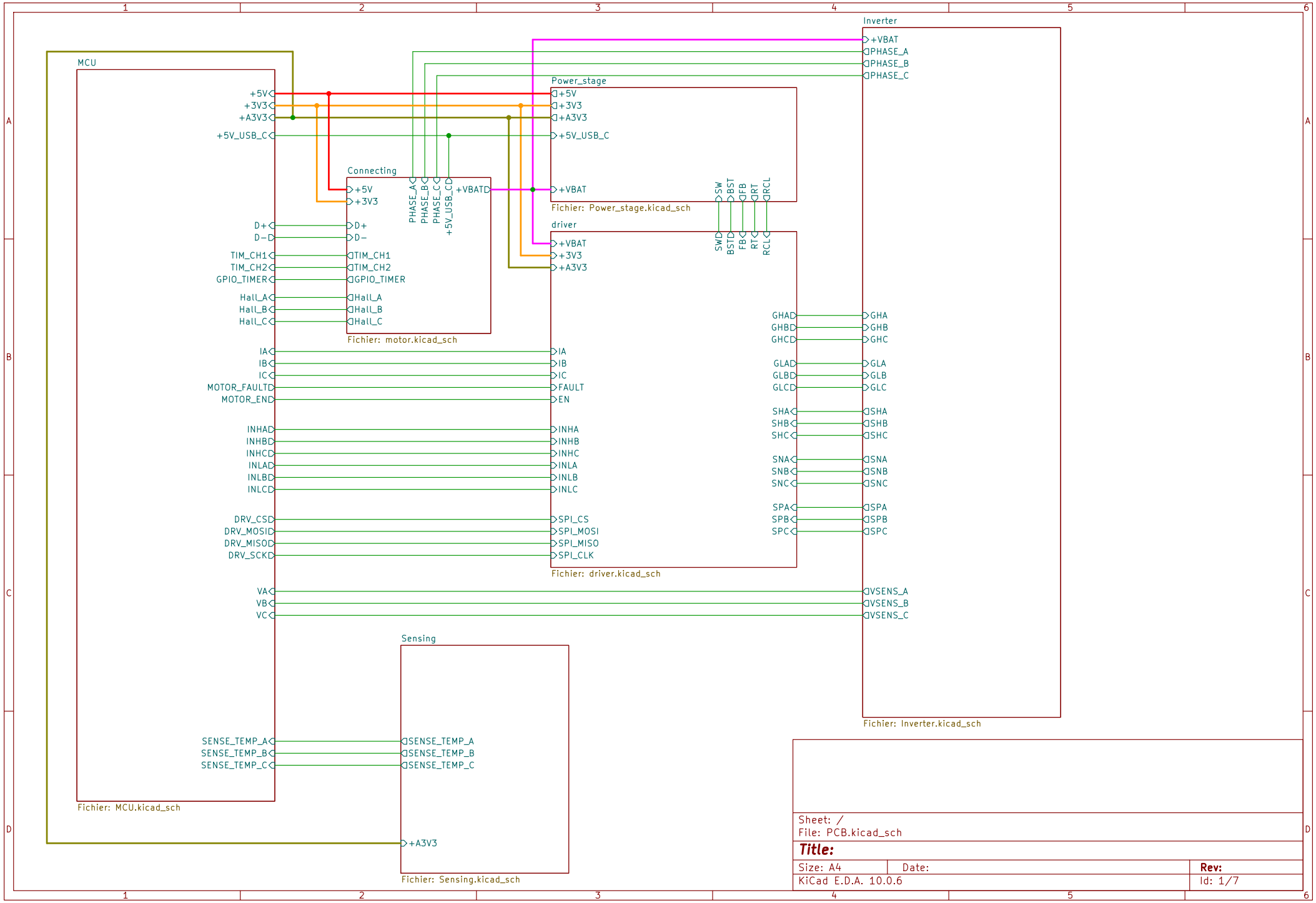
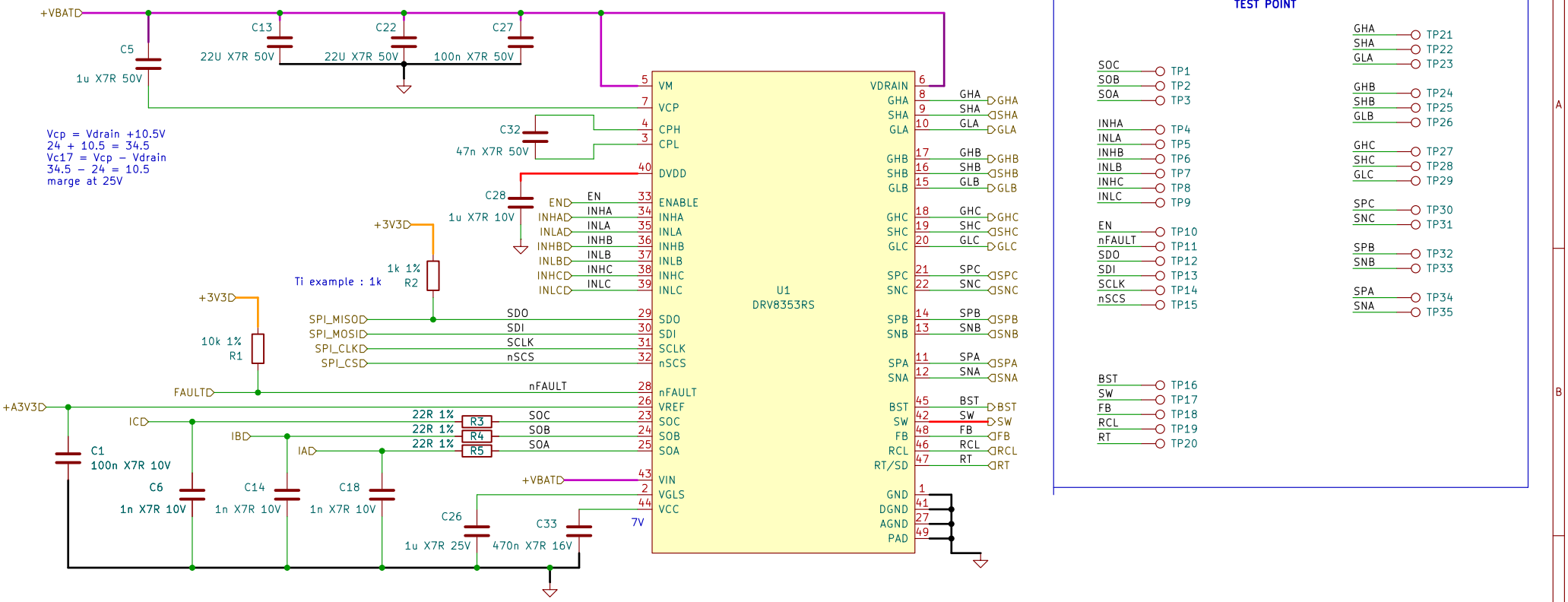
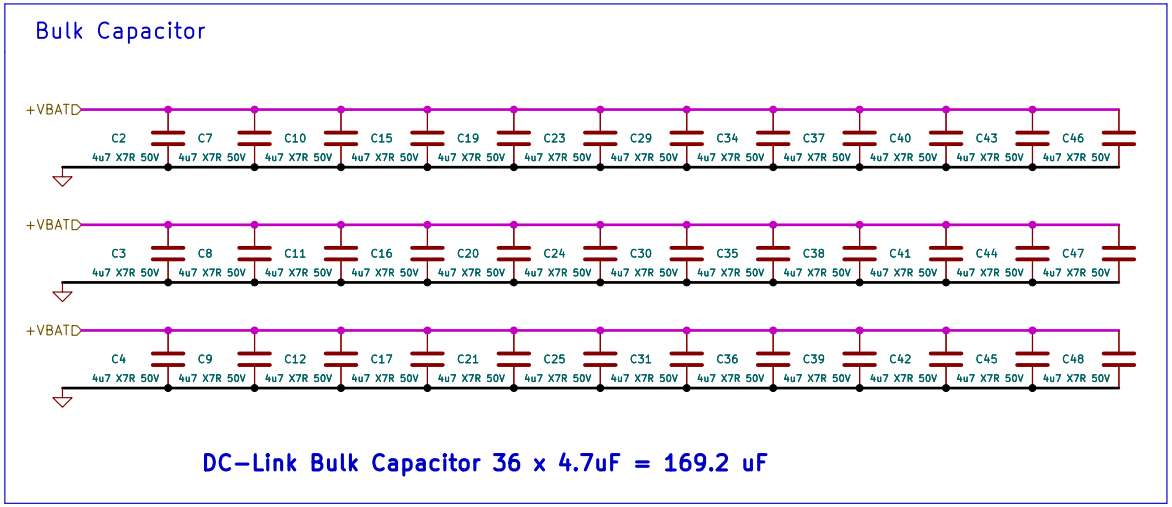

MCU
+5V
+3V3
+A3V3
+5V_USB_C
D+
D-
TIM_CH1
TIM_CH2
GPIO_TIMER
Hall_A
Hall_B
Hall_C
IA
IB
IC
MOTOR_FAULT
MOTOR_EN
INHAD
INHBD
INHCD
INLAD
INLBD
INLCD
DRV_CSD
DRV_MOSI
DRV_MISO
DRV_SCK
VA
VB
VC
SENSE_TEMP_A
SENSE_TEMP_B
SENSE_TEMP_C
Fichier: MCU.kicad_sch
Connecting
+5V
+3V3
PHASE_A
PHASE_B
PHASE_C
+5V_USB_C
+VBAT
D+
D-
TIM_CH1
TIM_CH2
GPIO_TIMER
Hall_A
Hall_B
Hall_C
Fichier: motor.kicad_sch
Power_stage
+5V
+3V3
+A3V3
+5V_USB_C
+VBAT
SW
BST
FB
RT
RCL
Fichier: Power_stage.kicad_sch
driver
+VBAT
+3V3
+A3V3
SWD
BSTD
FB
RT
RCL
IA
IB
IC
FAULT
EN
INHA
INHB
INHC
INLA
INLB
INLC
SPI_CS
SPI_MOSI
SPI_MISO
SPI_CLK
GHAD
GHBD
GHCD
GLAD
GLBD
GLCD
SHA
SHB
SHC
SNA
SNB
SNC
SPA
SPB
SPC
Fichier: driver.kicad_sch
Inverter
+VBAT
PHASE_A
PHASE_B
PHASE_C
GHA
GHB
GHC
GLA
GLB
GLC
SHA
SHB
SHC
SNA
SNB
SNC
SPA
SPB
SPC
VSENS_A
VSENS_B
VSENS_C
Fichier: Inverter.kicad_sch
Sensing
SENSE_TEMP_A
SENSE_TEMP_B
SENSE_TEMP_C
+A3V3
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Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 1/7

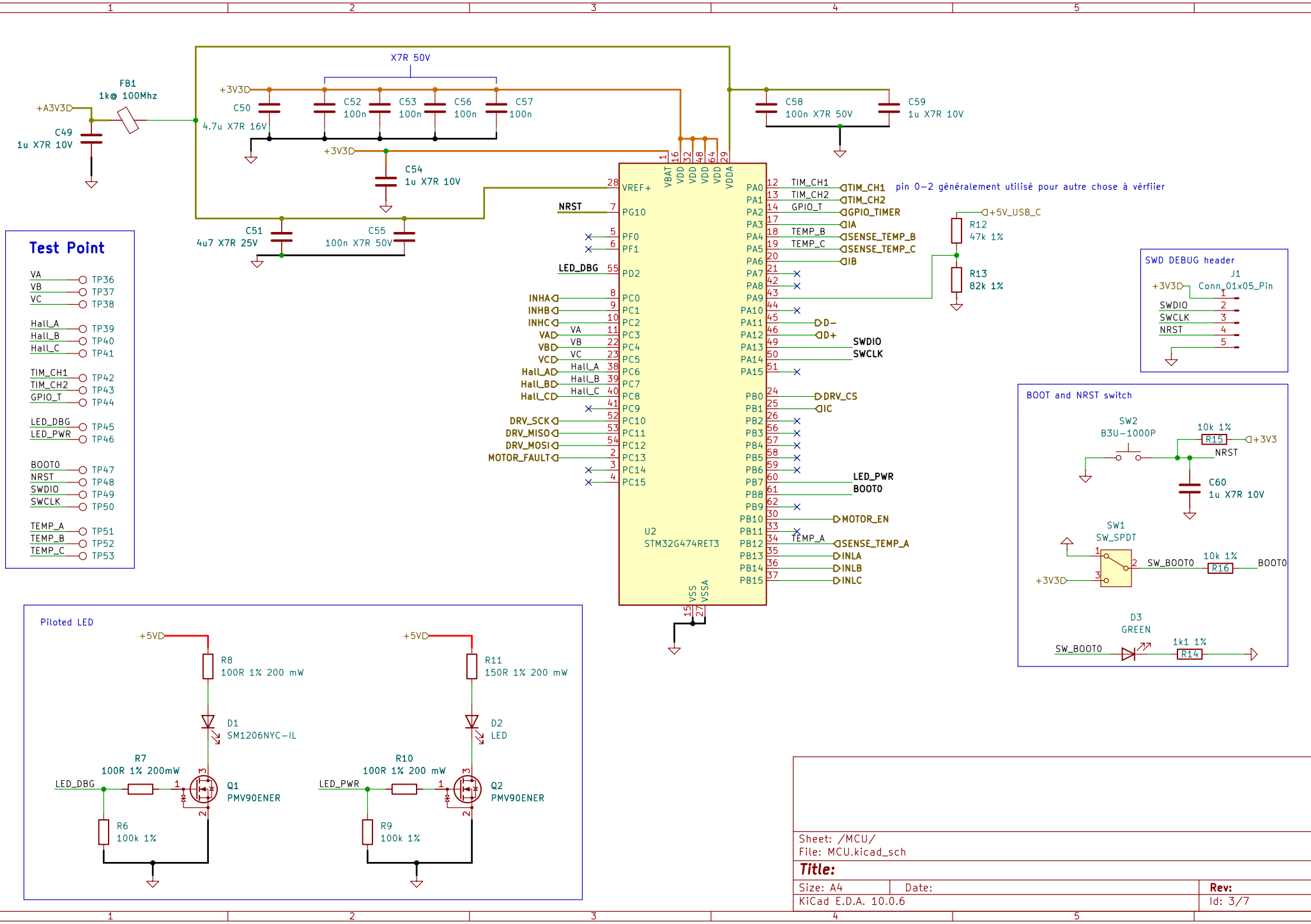




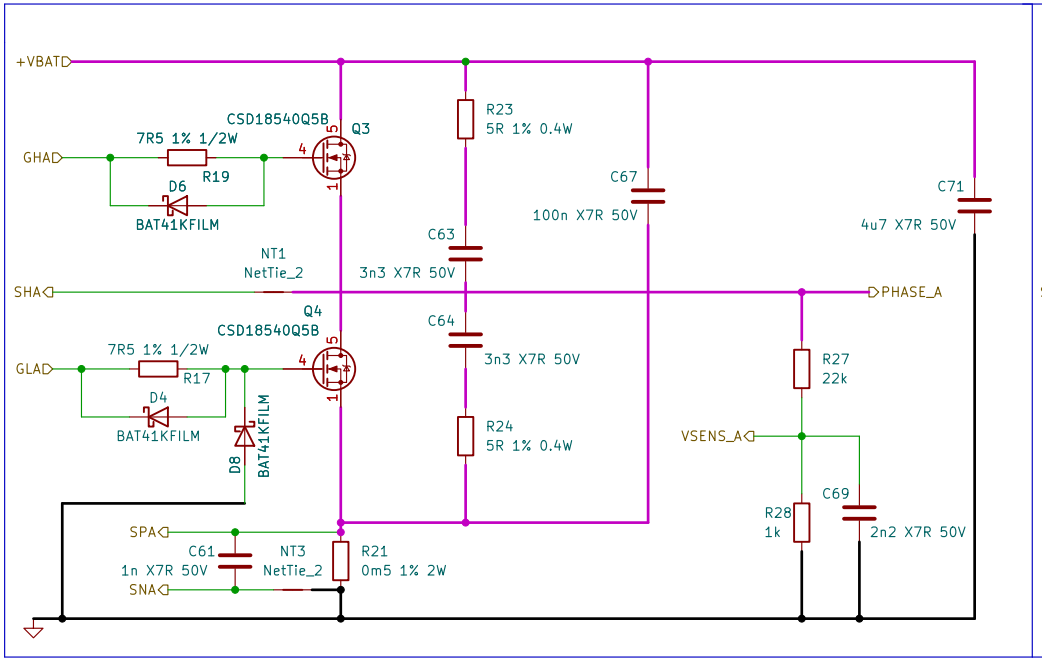
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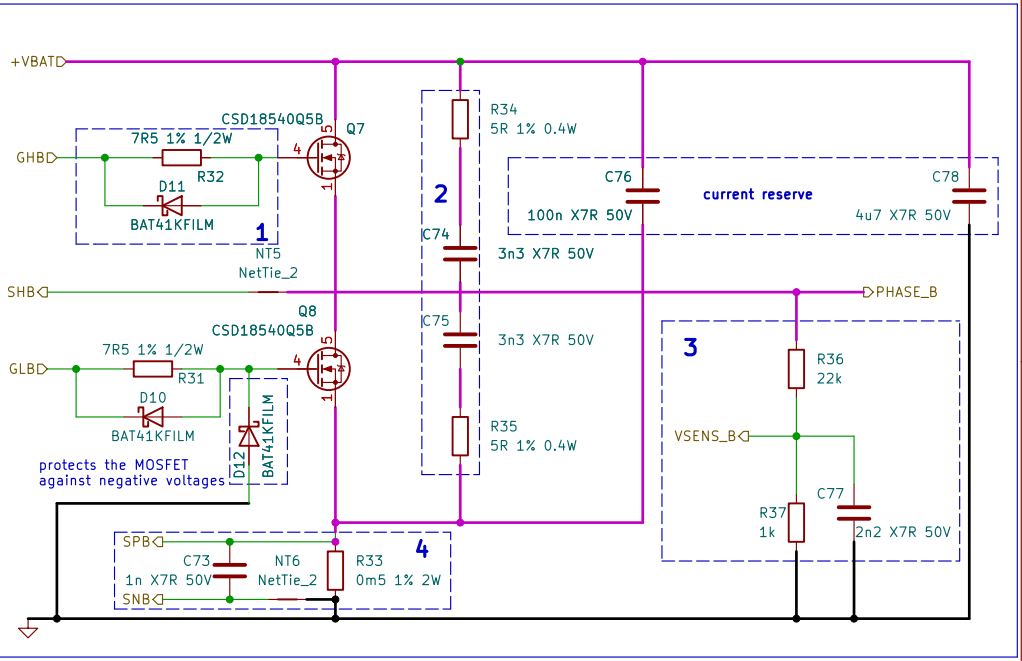
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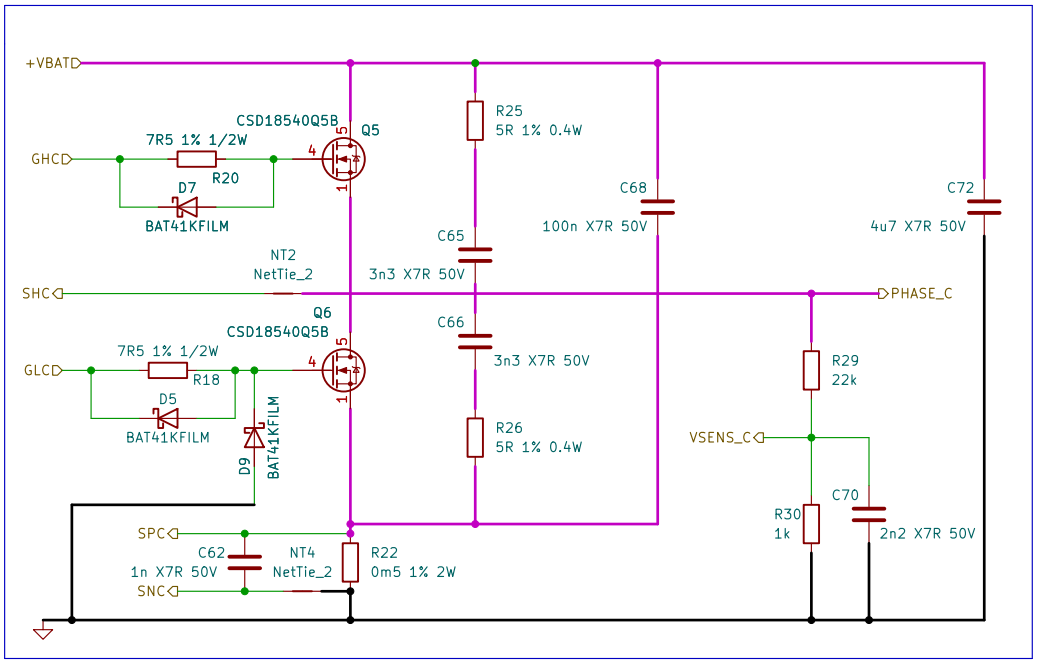
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/7



Sheet: /MCU/		
File: MCU.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/7







Mofset :
N = 6
Qg = 41 nC
Qgd = 6.7 nC
Rds(on),max = 2.2 mR
Imax = 100 A
Irms = I*sqrt(D)
Pcond = Irms**2 * Rds(on)
Pcond,max = 9.5 W
Psw = 0.5*Vbus*Isw(tr+tf)*fpwm

1

R : protect overtensions
+R better protection but slower
−R faster but less protection
tmiller = Qgd/Ig
e.g : G current = 200mA
tmiller = 33.5 ns

Diodes :
rapid discharge by short−circuiting the resistor

2

RC snuber :
attenuates the oscillations
tau = RC = 16.5 ns
fRC = 1/(2piRC) = 9.65 MHz
Pr = C(Vbus**2)fpwm

3

Vphase = 24 V
Vadc = Vphase/(Rhigh+Rlow) = 1.04V

4

R shunt :
current calculation
I = U/R, P = R*I**2
I = 15A −> Prshunt = 0.1125W
I = 30A −> Prshunt = 0.45W
I = 100A −> Prshunt = 5W

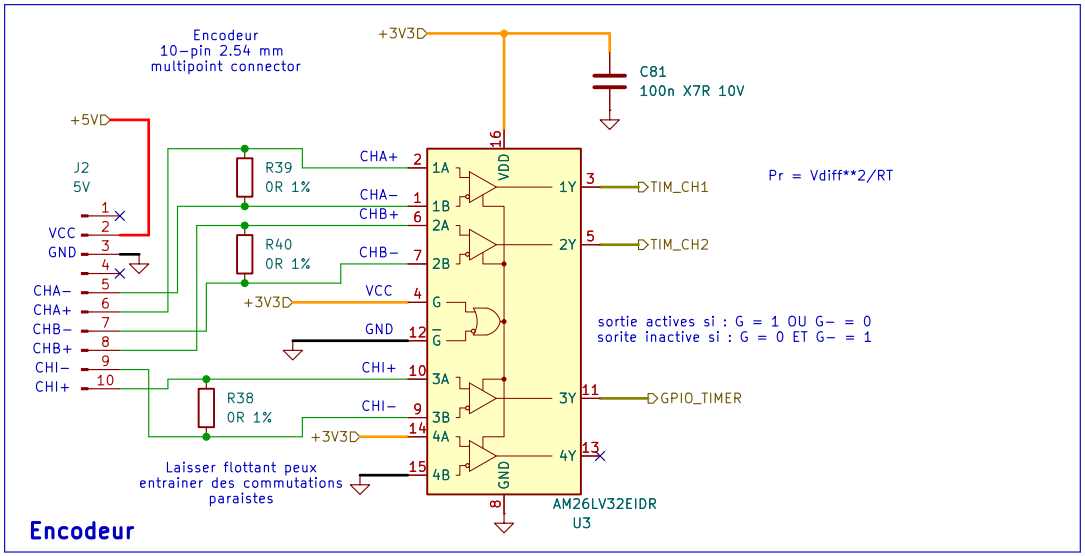
Vso = 1.65 + G*I*R
G = (5,10,20,40) (DRV 8353)
Ti recommand Vso = 0.25−3.05 V
G = 20 and I = 100 −> VSO = 2.65
less R −> high current

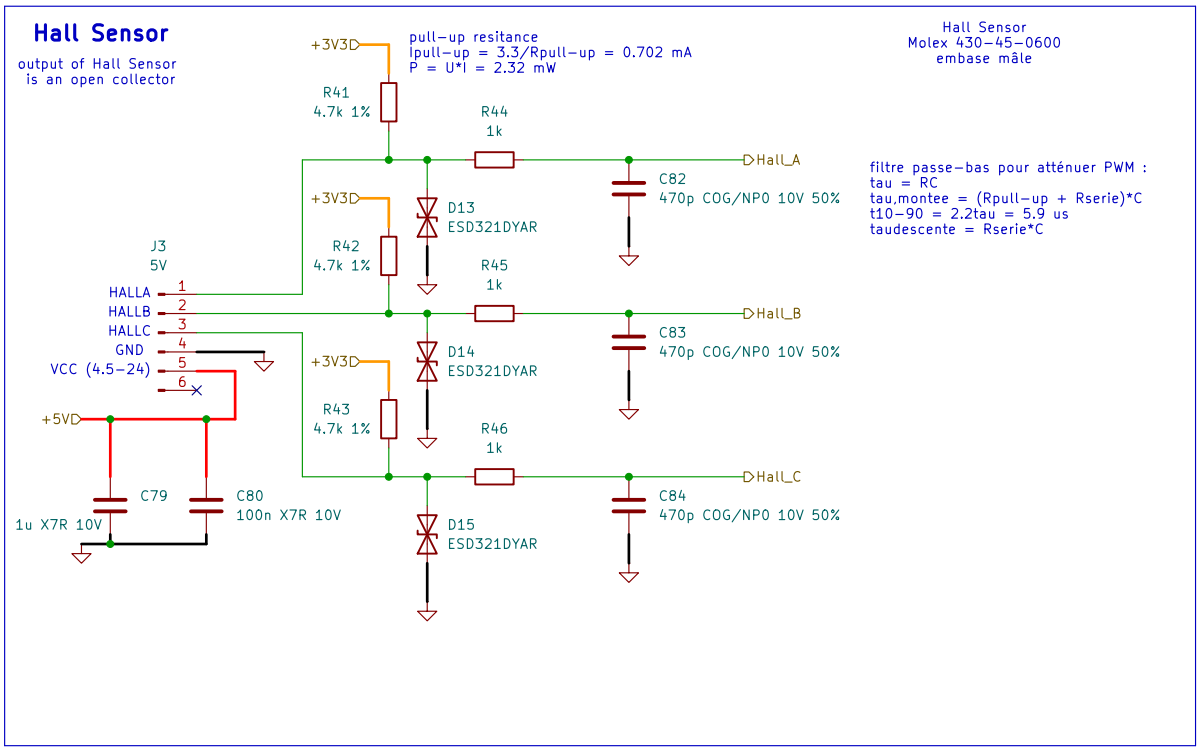
Cshunt :
attnuate noise

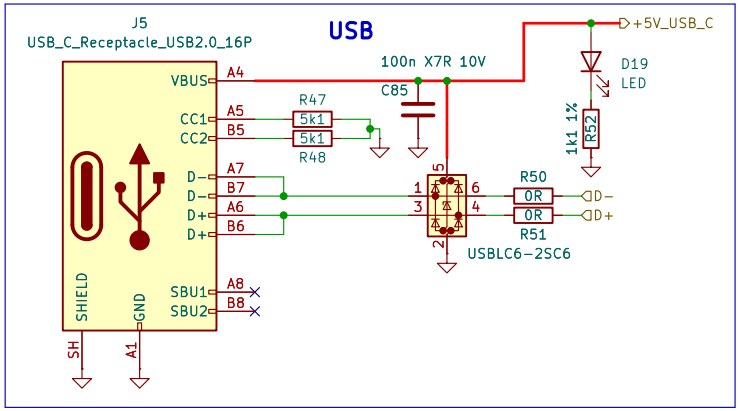
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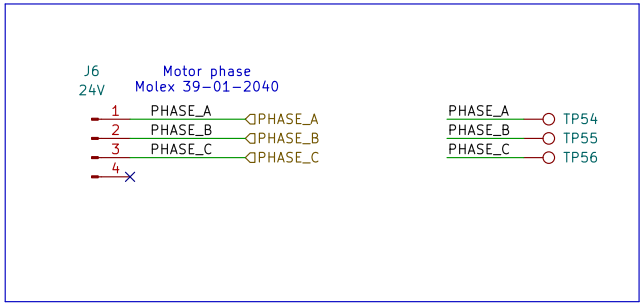
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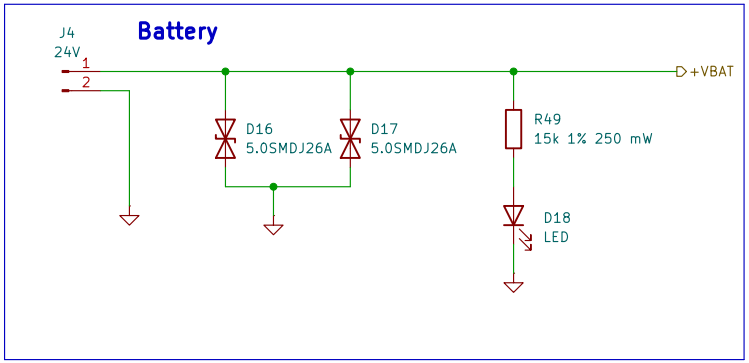
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KiCad E.D.A. 10.0.6 | Id: 4/7







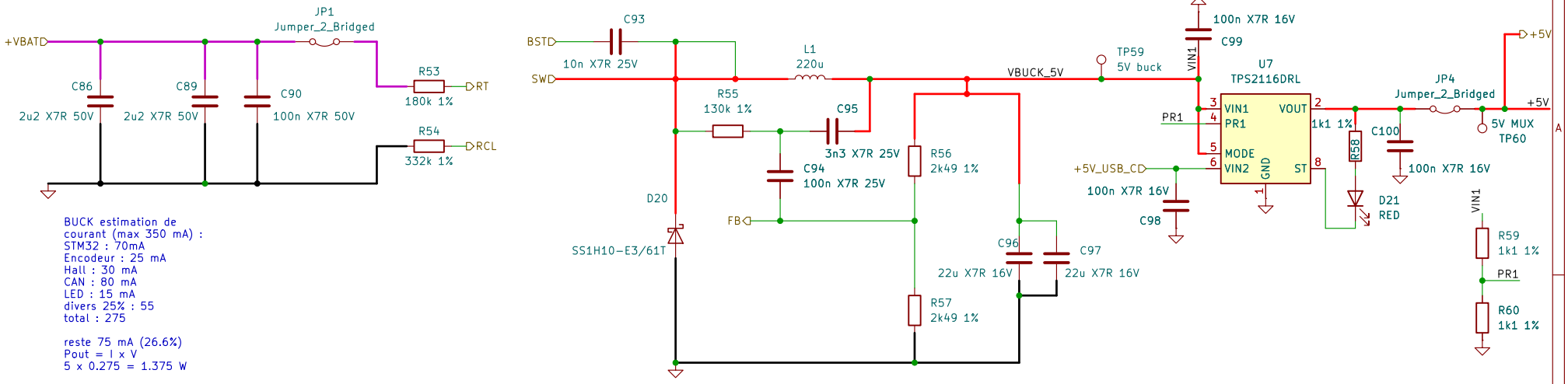




Sheet: /Connecting/
File: motor.kicad_sch

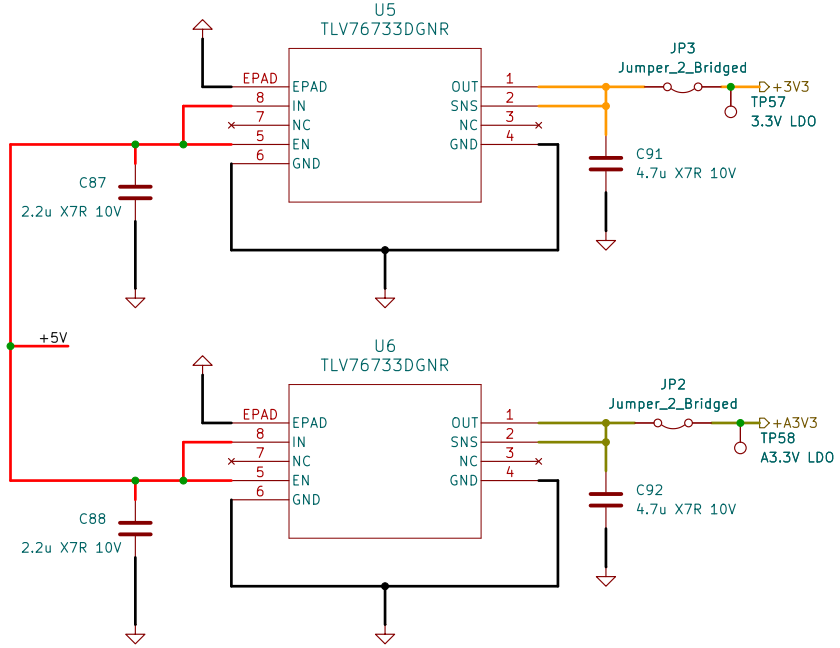
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/7



BUCK estimation de
courant (max 350 mA) :
STM32 : 70mA
Encodeur : 25 mA
Hall : 30 mA
CAN : 80 mA
LED : 15 mA
divers 25% : 55
total : 275

reste 75 mA (26.6%)
Pout = I x V
5 x 0.275 = 1.375 W



capacité d'entrée recommandé de 1 uF 2.2u
pour avoir une marge 10V 50% de marge

Capacitée de sortie Ti recommande 2.2uf 4.7 marge
sortie de 3v3 marge à 10V

Dimensionnement thermique :
PLDO = (Vin-Vout)*Iout
PLDO = (5-3.3)*Iout=1.7Iout
dT = RjA * PLDO
dT = 60.1 *1.7 *Iout
dT = 102.17*Iout

BUCK CONVERTER – DRV8353RS

Specifications:
Vin = 24 V, Vout = 5 V, Iout_max = 275 mA
Pout = Vout * Iout = 1.375 W
D = Vout/Vin = 0.208

Switching frequency – RT:
f = 200 kHz
(trade-off between efficiency and component size)
T = 1/f = 5 us
ton = D/f = 1.04 us > 400 ns (TI minimum)
RT = ton*Vin/(1.385e-10) = 180.5 kOhm
Selected RT = 180 kOhm (1%)

Inductor – L:
TI current limit: 410 mA min, 610 mA max
dIL < 2*(410-275) = 270 mA
Selected dIL = 90 mA (current margin)
L = (Vin-Vout)*ton/dIL = 220 uH
IL_peak = Iout+dIL/2 = 320 mA
Selected L = 220 uH, Isat > 610 mA
CCM for Iout > 45 mA, DCM below

Input capacitors – Cin:
Target input ripple = 0.5 V
Cin_min = Iout*ton/0.5 = 0.573 uF
Selected 2x 2.2 uF + 100 nF, X7R 50 V
Extra capacitance for DC bias and tolerances.

Output capacitors – Cout:
Target capacitive ripple < 5 mV
Cout_min = dIL/(8*f*0.005) = 11.25 uF
Selected 2x 22 uF, X7R 16 V
Low ESR, verify effective capacitance.

Feedback resistors:
Vout = 2.5*(1+Rtop/Rbottom)
Vout = 5 V -> Rtop = Rbottom
Selected 2x 2.49 kOhm (1%)

Ripple injection – RA, CA, CB:
Required for stable regulation with low-ESR capacitors.
VA = 5-1*(1-5/24) = 4.208 V
Target dVA = 40-50 mV (TI)
RA*CA = (Vin-VA)*ton/0.05 = 412 us
Selected RA = 130 kOhm, CA = 3.3 nF
RA*CA = 429 us -> dVA = 48 mV
CB = 100 nF (TI recommendation)
No series output ripple resistor required.

Current limit – RCL:
toff = T-ton = 3.96 us
toff_limit = (1.25*toff+0.35)*1.25 = 6.62 us
Calculated RCL = 321.4 kOhm
Selected RCL = 332 kOhm (1%)
Actual nominal toff_limit = 6.8 us

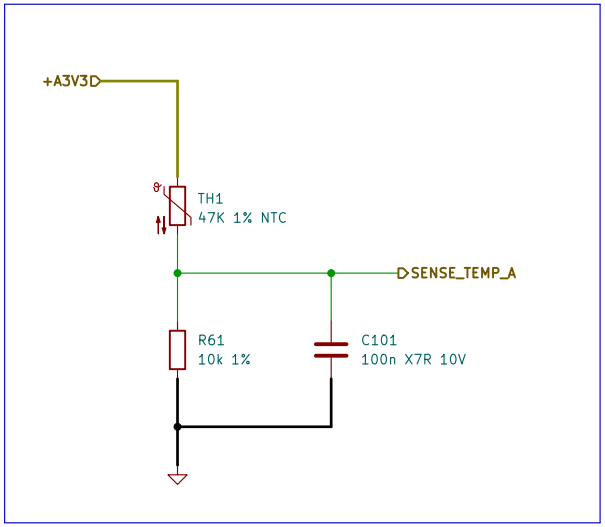
Other components:
CBST = 10 nF, X7R 25 V (BST-SW, TI recommendation)
CVCC = 1 uF, X7R 16 V (VCC-GND, TI minimum 0.47 uF effective)
Diode = Schottky, >=60 V, >=1 A
Anode = GND, Cathode = SW
Reverse voltage must exceed maximum bus voltage.

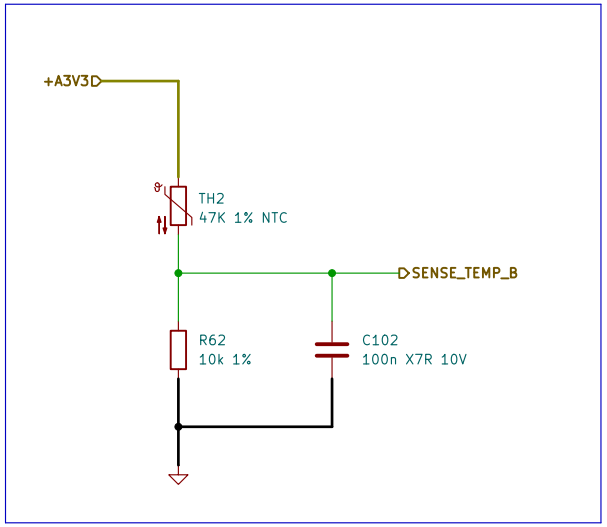
PCB considerations:
Keep VIN capacitor, diode and SW loop short.
Place BST and VCC capacitors close to DRV.
Keep FB away from SW and motor power traces.
Verify capacitor DC-bias, inductor saturation and thermal limits.

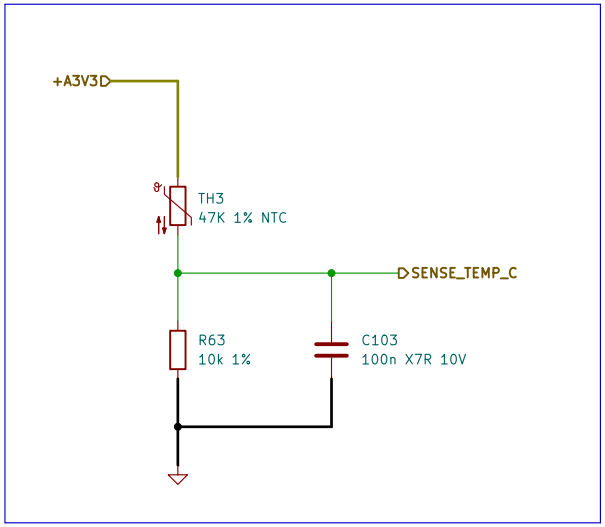
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/7







B = 44800, Rntc = 47 000, Tntc = 298.15 K
R(T) = Rntc * exp(B*(1/T − 1/Tntc)
Vadc = 3.3*R/(R(T)+R)

higer temp higer voltage ans lower R(T)

NOTE:
One temperature sensor must be physically placed between the high−side and low−side MOSFETs of each half−bridge.
This placement allows the sensor to estimate the average thermal conditions of both MOSFETs.
The temperature sensors are not electrically connected to the MOSFETs.

Sheet: /Sensing/
File: Sensing.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 7/7